SINGAPORE MANAGEMENT UNIVERSITY



OPIM 201 Operations Management

INVENTORY MANAGEMENT: THE NEWSVENDOR MODEL

(C&T, Ch. 14)

Motivation – Buying Newspapers



Order Newspapers from Supplier

Sell Random Number of Newspapers





Dispose of Unsold Newspapers

How many copies of newspapers should the newsvendor order?

The Newsvendor Model

- The newsvendor model considers a setting in which there is a single selling season and you have only one procurement opportunity.
  - This opportunity occurs well in advance of the selling season, and you receive your entire order before the selling season starts.
  - Fixed cost (e.g., ordering cost) is inconsequential. (Why?)
- Random demand occurs during the selling season
  - Excess inventory at end of season may be salvaged
- Applies to products with short life cycle
  - e.g., Newspaper/Magazines, Fashion industry, Seasonal products (Christmas trees, toys), Fresh food, Farming, Blood-banks
  - Typically long supply lead time
- Decision: ***Determine the order quantity, Q***
  - The "Too much/Too little problem":
    - Order too much and inventory is left over at the end of the season
    - Order too little and sales are lost

Thinking Broadly about Applications of Newsvendor Model

- The main features of the Newsvendor model are
  - An uncertain quantity (demand).
  - A bet on that quantity (made before the exact value of the uncertain quantity is revealed).
  - There are consequences of getting this bet wrong.
  - This structure extends to many situations beyond the traditional Newsvendor Setup.

An Uncertain Number	Ex-Ante Bet
Product Demand	*FGI/Stock Level*
# of Guests at a Party	*Amount of Food to order*
Talk Time Demand	*Cell-Phone Plan*
Market Size for new Product	*Capacity of Production Facility*
Project Complexity	*Number of employees on project*
Withdrawals at a hedge fund	*Cash reserve/Short Term Investments*

Building Intuition …



Order Newspapers
from Supplier

Sell Random
Number of Newspapers





Dispose of Unsold
Newspapers

- Economics:
  - Each newspaper sells for p = \$1.50
  - Supplier charges c = \$1 per newspaper
  - Salvage value for unsold newspapers = \$0.3
- Demand Forecast: Normally distributed with …
  - Average demand = 200 units per day
  - Standard deviation of demand = 30

Building Intuition …



Order Newspapers from Supplier

Sell Random Number of Newspapers





Dispose of Unsold Newspapers

- Economics:
  - Each newspaper sells for p = ~~$1.50~~ $5
  - Supplier charges c = $1 per newspaper
  - Salvage value for unsold newspapers = $0.3
- Demand Forecast: Normally distributed with …
  - Average demand = 200 units per day
  - Standard deviation of demand = ~~30~~ 60

Is a Point Forecast not Sufficient as Information?

	Scenario 1	Scenario 2	Scenario 3
Price	**$1.50**	**$5**	$1.50
Cost	$1	$1	$1
Salvage value	$0.30	$0.30	$0.30
Demand	Normal (200,**30**)	Normal (200,30)	Normal (200, **60**)
Order Quantity?			

- Would we order the same quantity for all three scenarios?
- The optimal order quantity depends on the risk (of demand uncertainty) and the economic benefits.

Notation

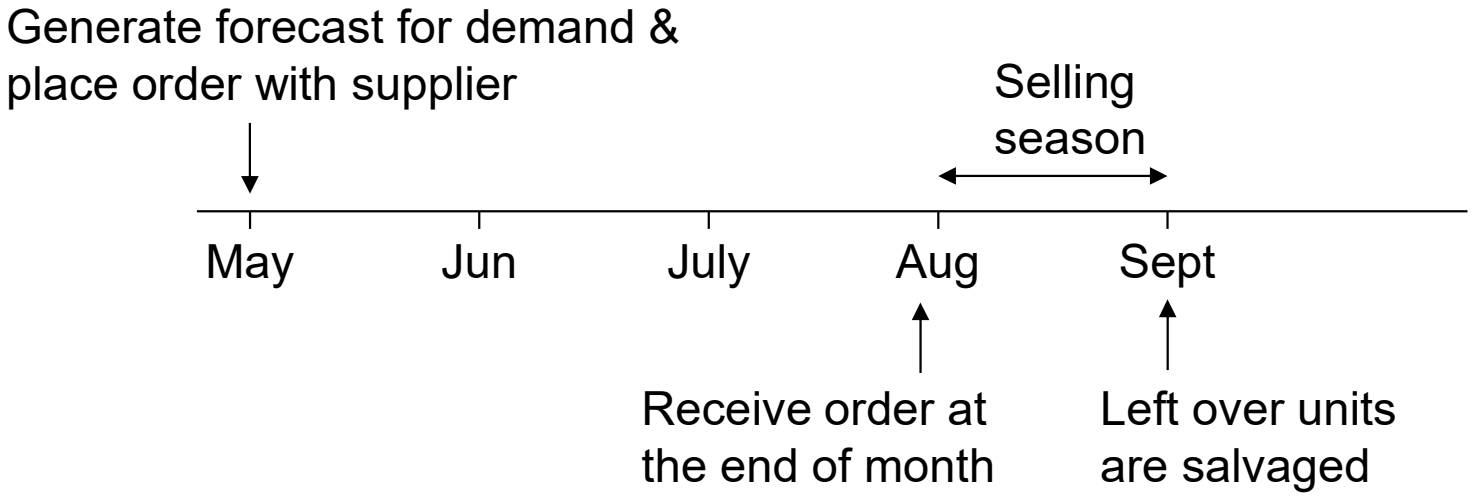
- Q — Order Quantity
- r — Unit revenue
- c — Unit (acquisition) cost
- s — Unit salvage value
- D — (Random) Demand
- F — Probability distribution function of demand, D
  - $F(x) = P(D \leq x)$

"Too Much" and "Too Little" Costs

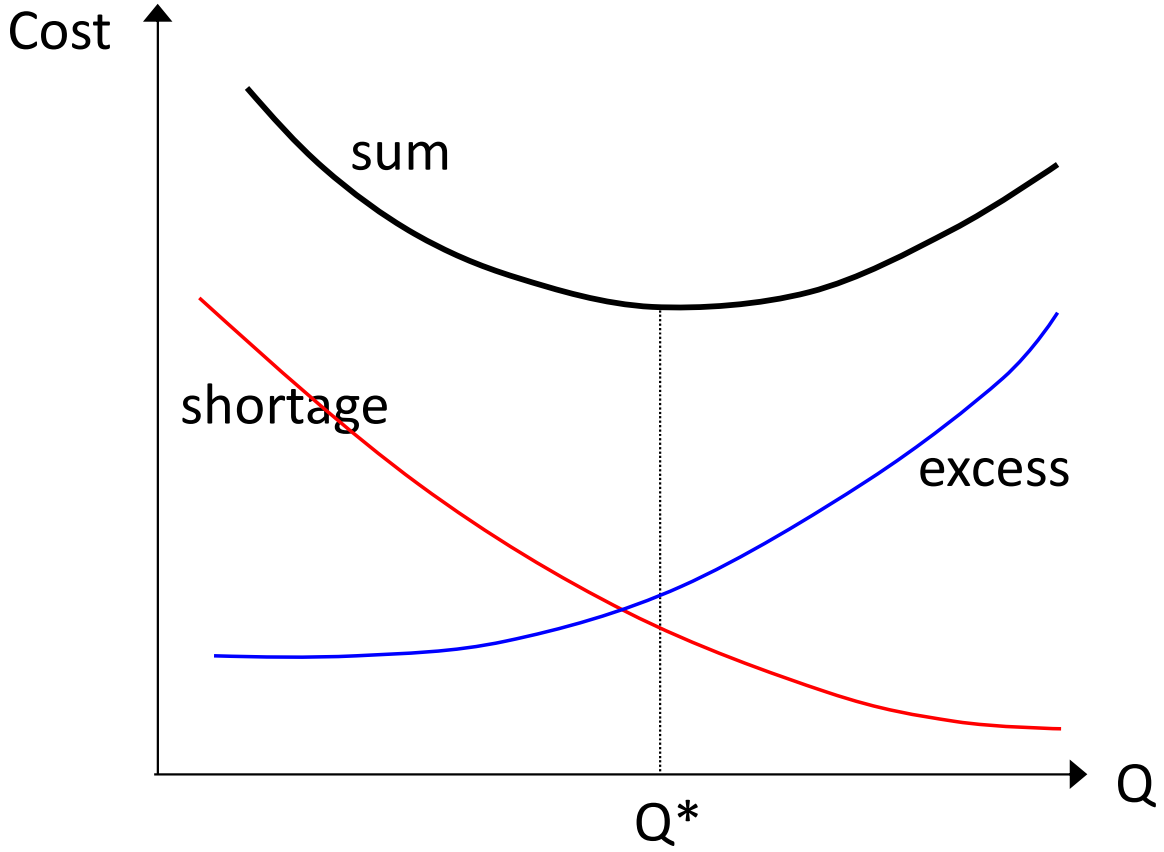
- Too much (Overage) cost, C_O
  - The loss incurred when a unit is ordered but not sold
  - In other words, suppose you had left over inventory (i.e., you over-ordered). Overage cost is the increase in profit you would have enjoyed if you had ordered one fewer unit.
    - For the newsvendor, Overage Cost = Purchase Cost – Salvage value
  - In general, it represents the unit cost of over-estimating demand.
- Too little (Underage) cost, C_U
  - The opportunity cost of not ordering a unit that could have been sold
  - In other words, suppose you had lost sales (i.e., you under-ordered). Underage cost is the increase in profit you would have enjoyed if you had ordered one more unit.
    - For the newsvendor, Underage Cost = Price – Purchase Cost
  - In general, it represents the unit cost of under-estimating demand.

Example 1 – The SMU Fashion Store

- The SMU Fashion Store sells a fashion item at the beginning of term 1. The store has to place an order for this item many months in advance of the fashion season in order to get a good price on the item. Each unit costs Fashion $10. These units are sold to customers at a price of $16 per unit. Units not sold during the season can be sold to the outlet store at $5 per unit. What is the order quantity that maximizes the profit to run the store?



Trade Off: Leftover Inventory Costs vs. Stock-out Costs



Objective: Finding optimal order quantity, Q* that minimizes inventory holding cost & stock-out cost

As Q ↑, the likelihood of excess inventory ↑, hence leftover inventory costs ↑; meanwhile the likelihood of shortages ↓, hence stock-out costs ↓.

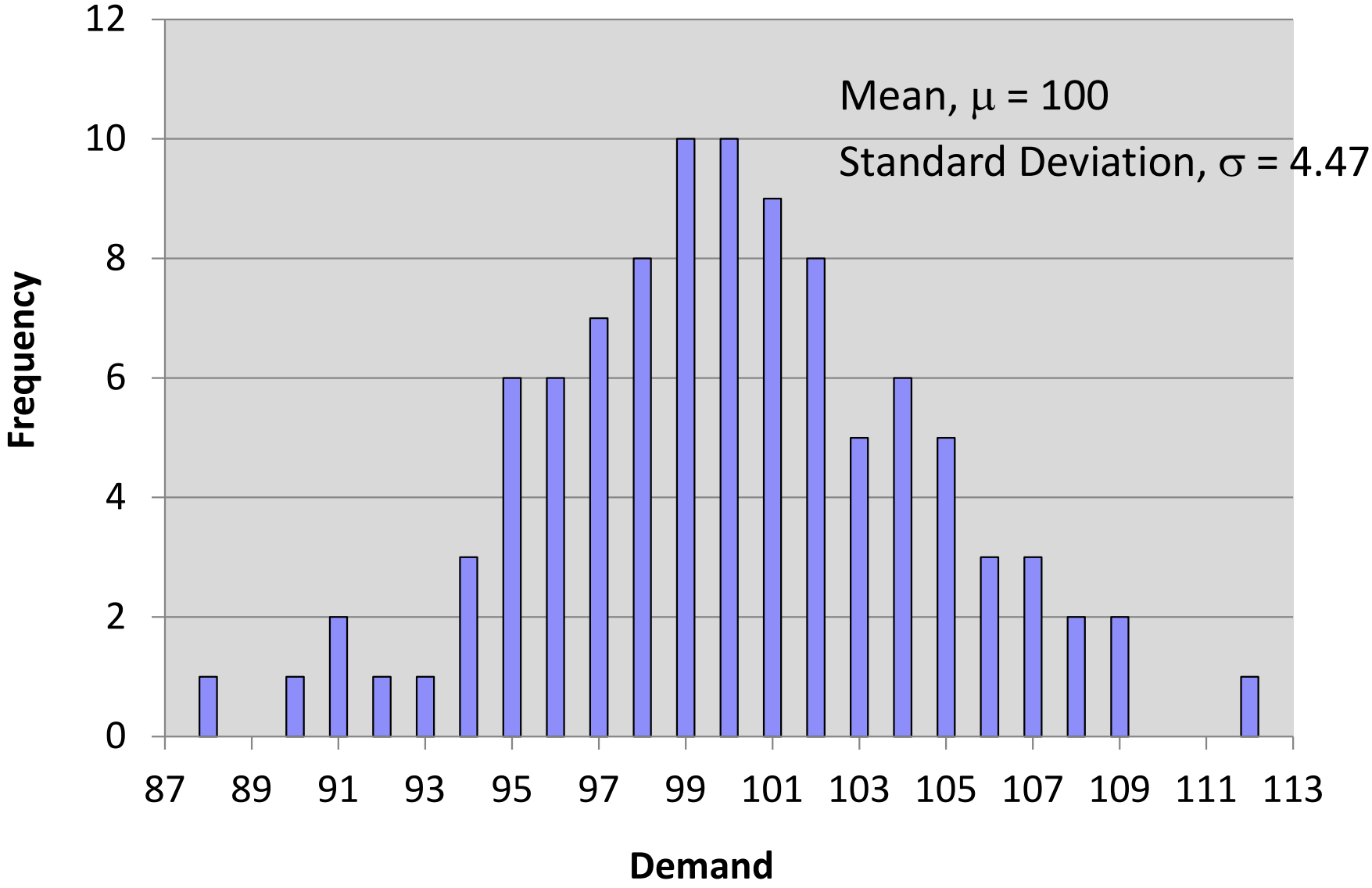
Trade Off: (Order) Too Much vs Too Little

Assessing Risk of Demand Uncertainty

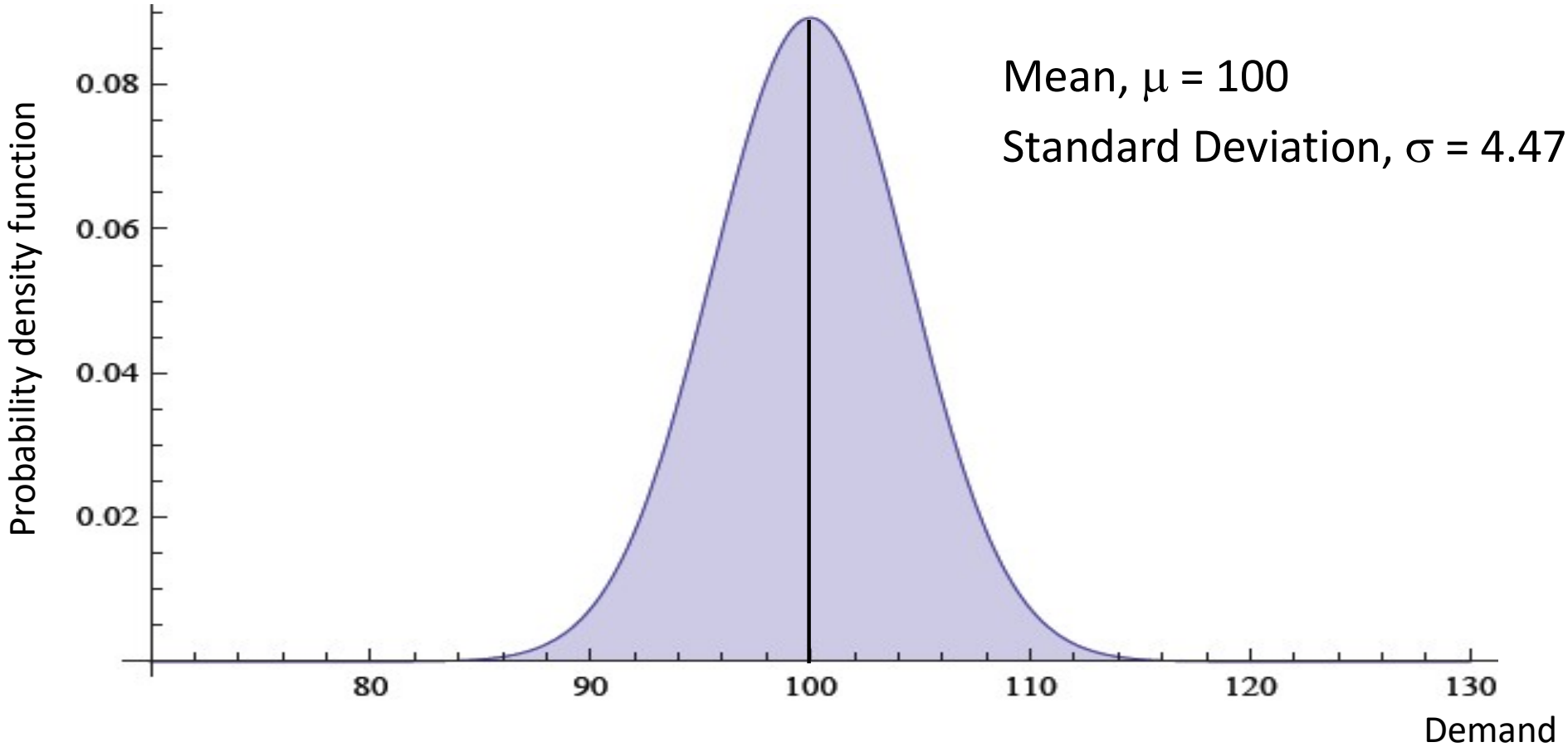
- The optimal order quantity depends on the risk of demand uncertainty. To assess this risk, we collect historical data from comparable items over the last few years and generate 100 demand observations as follows:

101	106	104	104	105	97	93	100	98	97	97	98	100	101	95
109	91	103	105	107	96	100	103	94	96	95	102	95	102	101
105	100	94	88	98	96	99	101	105	100	107	99	104	96	102
101	98	98	99	106	91	97	104	102	99	100	99	94	108	101
97	107	98	101	100	96	103	102	98	102	103	101	98	106	103
97	100	105	100	96	101	99	102	104	100	95	95	104	99	108
112	102	90	99	95	99	99	97	92	109					

Assessing Risk of Demand Uncertainty (cont'd)



Assessing Risk of Demand Uncertainty: Approximating Demand Distribution



The demand uncertainty can be characterized by approximating the demand distribution using a probability distribution function, denoted by F (e.g., Normal distribution)

Optimal Order Quantity

- For any continuous demand distribution function, F, the optimal order quantity, Q* that minimizes inventory holding cost and stock-out cost, satisfies

$$F(Q^*) = P(D \leq Q^*) = \frac{C_U}{C_U + C_O} \quad \sim \text{ Newsvendor Fractile}$$

- Here F(Q*) (= P(D $\leq$ Q*)) is the probability that demand is less than or equal to Q*. It may be interpreted as the ***In-stock Probability***, the probability that all demand is satisfied immediately from the primary stock (a measure of service level), so that the Newsvendor (Critical) fractile, $C_U/(C_U+C_O)$, may be interpreted as the *Optimal In-stock Probability*. Note that the larger C_U (C_O) is, the larger (smaller) the optimal in-stock probability.

- The ***Stock-out Probability*** , P(D >Q*) (= 1- F(Q*)), is the probability that some demand is lost.

Special Case: Normal Probability Distribution Function

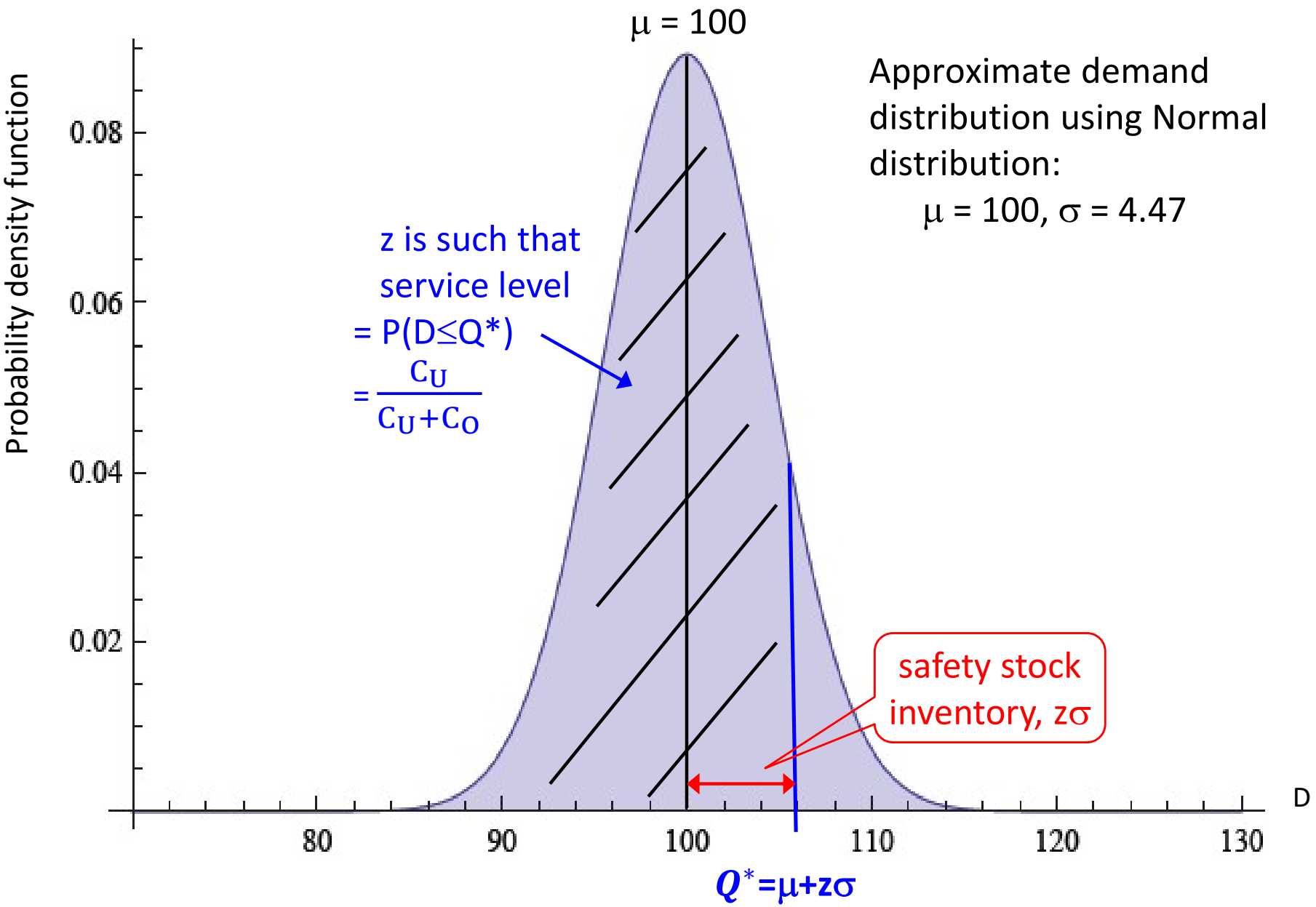
- When demand is Normally distributed with mean μ and standard deviation σ, i.e., $D \sim N(\mu, \sigma)$, then Q* simplifies to

$$Q^* = \mu + z \times \sigma$$

where $z = z_{Cu/(Cu+Co)}$ can be read from the Normal table for any given required service level $C_U/(C_U+C_O)$. Here $z \times \sigma$ can be interpreted as safety stock (to hedge against risk of demand uncertainty).

- Note that the safety stock $z \times \sigma$ (and hence Q*) depends on
  1) Degree of uncertainty (captured by σ)
  2) Target service level, $C_U/(C_U+C_O)$ (captured by z)
     - Trade-off of underage cost versus overage cost
- In practice, it is often challenging, if not impossible, to determine C_U and C_o. Very often, managers select a target service level intuitively and set the required safety stock (and Q*) accordingly.

Normal Probability Distribution Function (cont'd)



The SMU Fashion Store (cont'd)

C_O =

The cost of ordering one more unit than what you would have ordered had you known demand. In other words, suppose you had left over inventory (i.e., you over ordered). C_O is the increase in profit you would have enjoyed had you ordered one fewer unit.

C_U =

The cost of ordering one fewer unit than what you would have ordered had you known demand. In other words, suppose you had lost sales (i.e., you under ordered). C_U is the increase in profit you would have enjoyed had you ordered one more unit.

Hence $\frac{C_U}{C_U+C_O}$ =

From the Normal table, $z_{0.545} \approx 0.12$ (see next slide) so that

$$Q^* = 100 + 0.12 \times 4.47 = 100.54$$

Why is Q* so close to μ?

Looking up the Standard Normal Distribution Table, $\Phi(z)$

z	0.00	0.01	0.02	0.03	0.04	…	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595		0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	…	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	…	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	…	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	…	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540		0.71566	0.71904	0.72240
⋮		⋮	⋮					⋮	
⋮		⋮	⋮					⋮	
3.8	0.99993	0.99993	0.99993	0.99994	0.99994		0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	…	0.99996	0.99997	0.99997
4.0	0.99997	0.99997	0.99997	0.99997	0.99997		0.99998	0.99998	0.99998

- ***Round-up rule***: If the critical ratio falls between two values in the table, choose the entry that leads to the *greater* order quantity.
- In this case, since $0.54380 < \frac{C_U}{C_U+C_O} = 0.545 < 0.54776$, we choose F(z) = 0.54776.
- Hence $z_{0.545} \approx 0.12$.

Looking up the Standard Normal Distribution Table, $\Phi(z)$

z	-0.09	-0.08	-0.07	-0.06	-0.05	…	-0.02	-0.01	0.00
-4.0	0.00002	0.00002	0.00002	0.00002	0.00003		0.00003	0.00003	0.00003
-3.9	0.00003	0.00003	0.00004	0.00004	0.00004	…	0.00004	0.00005	0.00005
-3.8	0.00005	0.00005	0.00005	0.00006	0.00006	…	0.00007	0.00007	0.00007
-3.7	0.00008	0.00008	0.00008	0.00008	0.00009	…	0.00010	0.00010	0.00011
-3.6	0.00011	0.00012	0.00012	0.00013	0.00013	…	0.00015	0.00015	0.00016
-3.5	0.00017	0.00017	0.00018	0.00019	0.00019		0.00022	0.00022	0.00023
⋮		⋮	⋮					⋮	
⋮		⋮	⋮					⋮	
-0.2	0.38591	0.38974	0.39358	0.39743	0.40129		0.41294	0.41683	0.42074
-0.1	0.42465	0.42858	0.43251	0.43644	0.44038	…	0.45224	0.45620	0.46017
0.0	0.46414	0.46812	0.47210	0.47608	0.48006		0.49202	0.49601	0.50000

- The z-values can be negative, in which case the service level is less than 50%.
  - Note that $\frac{C_U}{C_U+C_O} < 0.5$ if $C_U < C_O$

Newsvendor Model Implementation Steps

- Gather economic inputs:
  - Selling price, production/procurement cost, salvage value of inventory
- Generate a demand model:
  - Use empirical (discrete) demand distribution (see next slide) or choose a standard distribution function to represent demand, e.g. the normal distribution
- Choose an objective:
  - e.g., maximize expected profit (i.e., satisfy the optimal service level) or satisfy a service level constraint (i.e., satisfy a target service level)
- Choose a quantity to order

Discrete Distribution Function

d	Frequency	P(D = d)	P(D ≤ d)
88	1	0.01	0.01
90	1	0.01	0.02
91	2	0.02	0.04
92	1	0.01	0.05
93	1	0.01	0.06
94	3	0.03	0.09
95	6	0.06	0.15
96	6	0.06	0.21
97	7	0.07	0.28
98	8	0.08	0.36
99	10	0.1	0.46
100	10	0.1	0.56
101	9	0.09	0.65
102	8	0.08	0.73
103	5	0.05	0.78
104	6	0.06	0.84
105	5	0.05	0.89
106	3	0.03	0.92
107	3	0.03	0.95
108	2	0.02	0.97
109	2	0.02	0.99
112	1	0.01	1

In-stock probability

Suppose we use the original data (frequency table) instead of the Normal approximation as the demand distribution.

$$\frac{C_U}{C_U + C_O} = 0.545$$

Q*?

Betting on Uncertain Demand – Insights

- Losses (mismatch cost) incurred even at "optimal" order quantities; methods to improve this situation:
  - Improve forecast accuracy
  - Influence demand during season (dynamic pricing, promotions)
  - Find reactive supply (even if more expensive) to reduce underage costs; instead of losing the margin, you pay a little extra to make the sale
  - Reduce overage costs: discount outlets, transshipment, remanufacturing, recycling
- Optimal ordering can be counterintuitive – cost structure matters!
  - Ordering average demand (which is usually the forecast) is only optimal when cost of overage equals cost of underage (when critical ratio = 0.5)
  - Optimal order quantity can be less than forecast (if cost of overage > cost of underage)
  - Products with same forecasts may not have same order quantities;
  - So, always check your intuition by using the newsvendor model!

Newsvendor Model Application: Revenue Management

- Revenue management is a technique to maximize the revenue earned from a fixed, perishable resource (e.g. airline seats, hotel room, concert tickets) by matching fixed supply with uncertain demand.
- This is achieved by influencing demand (as opposed to forecasting) through
  - Price discrimination
  - Capacity control.

- Examples:







Revenue Management: Common Features

- Fixed amount of resources or products or service to sell
- Finite selling season beyond which value of the product or service drops
- Fixed costs are sunk, marginal costs are small, thus the focus is on revenue
  - Hence the name “revenue” management
- Demand is uncertain
- Customers are heterogeneous
  - Typically more valuable customers arrive later
  - Hence price discrimination is used

Example 2: Hotel Management

- You manage a hotel with 200 rooms in Singapore
  - F1 is coming up
  - F1 fans either book a room in advance or wait until the last minute
  - Advance booking (bargain rate) vs. late booking (premium rate)
- Suppose you can easily get 200 advance booking and fill up your hotel. Should you fill up rooms with advance customers?

Room Reservation	Newsvendor Problem
Hotel rooms are "perishable"	Newspapers are "perishable"
Random last-minute demand	Random demand
Decide how many rooms to reserve for last-minute customers	Decide how many newspapers to purchase
If reserve too few rooms, miss potential premium customers	If stock too few newspapers, miss potential sales
If reserve too many rooms, revenue lost on empty rooms	If stock too many newspapers, money wasted on unsold newspapers
Bargain rate (advance)	Unit cost of newspapers
Premium rate (last-minute)	Unit selling price

Example 3: Airline Overbooking Problem

Next week, Super Discount Airlines has a flight from New York to Los Angeles that will be booked to capacity. The airline knows from its past history that an average of 25 customers (with a standard deviation of 15) cancel their reservation or do not show up for the flight. Revenue from a ticket on the flight is $125. If the flight is overbooked, the airline has a policy of getting the customer on the next available flight and giving the person a free round-trip ticket on a future flight. (To simplify the problem, assume that the customer would have booked the next available flight himself had the airline not accepted his original booking.) The cost of this free round-trip ticket averages $250. Super Discount considers the cost of flying the plane from New York to Los Angeles a sunk cost. Assume the demand follows a Normal distribution. By how many seats should Super Discount overbook the flight?

Example 3: Airline Overbooking Problem – Solution

- What is demand? What is uncertain here?
  "Demand" is the cancellations, i.e., we need to estimate # of cancellations.

- What is the decision that SIA needs to make? (Suppose total # of available seats =100)
  - Total number of tickets to sell?
  - Number of tickets to sell beyond 100?
  - $100 + Q$, where Q is the number of tickets sold beyond capacity

- C_U = Cost of under-estimating demand (# of no-show)
  - Suppose we estimate # of no-show to be 10; we overbook by 10 and sell a total of 110 tickets.
  - Suppose actual # of no-show is 11, i.e., we underestimate "demand".
    - 99 passengers show up, i.e., aircraft takes off with one empty seat
    -

Example 3: Airline Overbooking Problem – Solution

- C_O = Cost of over-estimating demand (# of no-show)
  - Suppose we estimate # of no-show to be 10 but the actual # of no-show is 9, i.e., we overestimate "demand".
    - 101 passengers show up for the flight
    -

Airline Overbooking Problem

Trade-Off:

Cost of Not Having a Date

Vs.

Loss of Goodwill



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